IEEE ROBOTICS & AUTOMATION SOCIETY
IUT STUDENT BRANCH CHAPTER

iTest BANGLADESH

RULEBOOK

iTest BANGLADESH Presents

PCB DESIGN CONTEST

Technical Partner EasyEDA

IEEE ROBOTICS & AUTOMATION SOCIETY
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iTest BANGLADESH

1. Participant Eligibility

- The competition is open to all current undergraduate students from any recognized university in Bangladesh.
- The competition will be conducted as an individual competition.

2. Competition Details

- The competition will consist of two rounds.
- Both rounds will be conducted online.
- For **Round 01**, the questions/problems will be provided **60 hours** before the submission deadline.
- For **Round 02**, the questions/problems will be provided **48 hours** before the submission deadline.
- No submission will be accepted after the specified deadline.
- Submission in both rounds is mandatory for consideration in the final merit list.
- Participants ***who submit in only Round 01 will not be considered for the final merit list.***
- Participants need to submit ***at least one problem fully to qualify for Round 02***. Participants who skip Round 01 will not be eligible to participate in Round 02.

2. Competition Details

- ***30% of the marks obtained in Round 01*** will be added to the final competition score.

3. Allowed Software

Participants are allowed to use **EasyEDA only** for designing their PCBs.

4. PCB Design Rules

4.1 Component and Pin Labelling

- Participants must label each component as well as the pins on the PCB.
- Components should be labelled using their respective component names. Examples: LCD, Arduino, Potentiometer, Timer.
- Pins must be labelled using the same names used in the main schematic or circuit diagram. Examples: VCC, GND, RS, RW, A0, A1.

4.2 Component Placement and Soldering

- Participants should design the PCB with the soldering process in mind.
- If a design requires soldering on both sides of the PCB, participants should modify the design as much as possible so that soldering is required on only one side.

4.3 Component Selection

- If the exact component specified in the schematic cannot be identified, participants may use a similar or equivalent component.
- Both SMD and Through-Hole (THT) components are permitted, depending on the participant's preference and the requirements of the problem statement.
- SMD components will be given higher priority during evaluation.

4.4 PCB Design Specifications

- The following minimum design specifications must be maintained:

Parameter	Requirement
Clearance	Minimum 0.3 mm
Trace Width	Minimum 0.25 mm
Via Diameter	Minimum 0.35 mm
Drill Diameter	Minimum 0.25 mm

Clearance requirements apply to:

- Pad-to-pad
- Pad-to-trace
- Trace-to-trace
- Trace-to-board edge

Note: It is recommended to use different and appropriate trace widths for power and signal traces.

5. Submission Guidelines

A Google Classroom will be provided for submission. The classroom link and access code will be sent to registered participants via email.

Participants must submit the following files:

5.1 Main Project Files

- The complete Schematic file
- The complete PCB Board file

5.2 Layer PDF

- A PDF containing all PCB layers must be submitted.
- Each layer should be placed on a separate page.
- Appropriate text identifying the corresponding layer must be added to each page.

5.3 Comprehensive Report

- A comprehensive report on the PCB design must be submitted.
- The report must include:
- 2D view of the PCB design
- 3D view of the PCB design

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5.4 File Submission Requirements

- All required files must be submitted individually.
- ZIP files are prohibited.
- File name must follow name_uni.filetype (e.g., Sadi_IUT.sch/.pdf/.json). **Any other format will not be considered for evaluation.**

6. Judgment Criteria

The PCB designs will be evaluated based on the following criteria:

6.1 DRC Compliance

- The design must comply with the specified Design Rule Check (DRC) requirements.
- 5 marks will be deducted for every DRC error.

6.2 Design Optimization

Bonus marks may be awarded for effective optimization of the PCB design.

The following factors will be considered:

- Board size optimization
- Number of via
    1. Fewer vias are preferred.
    2. Exceptions may apply to stitched vias.

- Component placement
- Interference tolerance
- Consideration of:
    1. Heating issues
    2. Power spikes
    3. Efficient component separation
- Overall aesthetics and appearance

7. Code of Conduct

Participants are expected to conduct themselves professionally and respectfully throughout the competition.
Any form of misconduct, including but not limited to:

- Cheating
- Disruptive behaviour
- Violation of competition rules

may result in immediate disqualification.

8. Event Information

For more details, visit the event page for:
iTest Banglaadesh Ltd. Presents PCB Designing Workshop and Contest- Powered by EasyEDA.

9. Contact Information

Participants may contact the following organizers directly for any queries regarding the competition.

Fariha Rahman Sneha
Email: fariha20@iut-dhaka.edu
Phone: 01302395583

Shoaib Mahbub
Email: shoaibmahbub@iut-dhaka.edu
Phone: 01866953000

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10. Competition Timeline

- **Round 1:** 4 October, 12:01 AM → 6 October, 11:59 AM (60 hours)
- **Round 2:** 8 October, 12:01 AM → 9 October, 11:59 PM (48 hours)

Please note the AM/PM timings carefully to avoid confusion.

11. Disclaimer

- The organizers reserve the right to make changes to any of the above rules and regulations to ensure that all competitors have an enjoyable competition.
- In the event of any ambiguity or uncertainty regarding these rules, the organizers' interpretation of the relevant clause shall prevail.

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